PRECALCULUS · UNIT 4: TRIGONOMETRIC FUNCTIONS

Daylight and Seasons

Model hours of daylight and temperatures with sinusoidal functions.

CP1

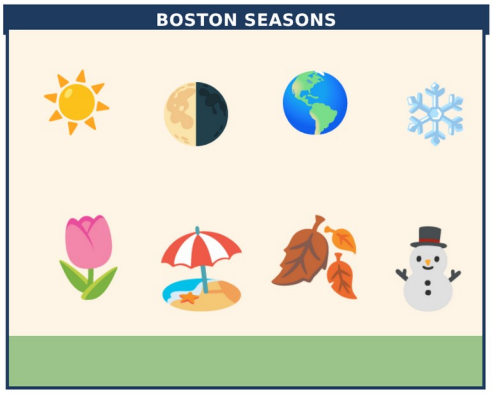
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The parks department wants to schedule outdoor events when there is plenty of daylight and comfortable weather. Daylight and temperature rise and fall every year in a repeating pattern.

You will use the unit circle and sinusoidal functions to model the seasons and recommend event dates.



Longest Day
June 21: 15.3 h

Shortest Day
Dec 21: 9.1 h

Learning Goals

- I can use radians and the unit circle to evaluate trig functions.
- I can build sinusoidal models with amplitude, period, phase shift, and midline.
- I can solve trigonometric equations in context.

Project Timeline

When	What you will do
Day 1	Unit Circle
Day 2	Build the Models
Day 3	Solve for Dates
Final	Seasonal Planning Report + reflection

Part 1: Unit Circle

Day 1 · Use radians.

Boston Data Cards

Max Daylight	Min Daylight	Period
15.3 h	9.1 h	365 days
day 172 (June 21)	day 355 (Dec 21)	

Avg Temp High	Avg Temp Low	Event Rule
74°F in late July	29°F in late Jan	Need at least
(day 205)	(day 22)	13 h of daylight

1. Through what angle in radians does Earth travel in 73 days?

2. Find the exact values of $\sin\frac{2\pi}{3}$, $\cos\frac{7\pi}{6}$, and $\tan\frac{5\pi}{4}$.

3. A point on Earth's orbit (radius 1 AU) is at angle $\frac{5\pi}{3}$. Find its coordinates.

4. Explain why $\sin\theta = \sin(\pi - \theta)$ using the unit circle.

Part 2: Build the Models

Day 2 · Use the Boston Data Cards.

5. Write a cosine model for daylight with its maximum on day 172.

6. Write a cosine model for average high temperature (max 74°F on day 205, min 29°F).

7. Find D(100) and T(100).

8. Why do the daylight and temperature models have different phase shifts?

Part 3: Solve for Dates

Day 3 · Need at least 13 h of daylight and a temperature of at least 60°F.

9. Solve D(t) = 13 for the two days in the year.

10. Solve T(t) = 60 for the two days.

11. Find the event window where both conditions hold, and give it as calendar dates.

Final Product: Seasonal Planning Report

Your final product must include:

- Unit-circle work
- Daylight and temperature models
- Predictions
- Solutions of trig equations
- Event window recommendation

Reflection

R1. How did you find the phase shift from the data?

R2. Why does a trig equation have two solutions in a year?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Sinusoidal models are built from data and used correctly to predict and solve for event dates. Goes beyond the minimum with a creative, realistic choice.	Sinusoidal models are built from data and used correctly to predict and solve for event dates.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes models, graphs, solutions, and a clear recommendation. Polished, organized, and visually engaging.	Report includes models, graphs, solutions, and a clear recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.